

How Much Does Air Connectivity Increase Aviation CO2? ✈️

Transportation and Energy Economics

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Brief Introduction about Me

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CURRENT

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Kyushu University (Faculty of Engineering)
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RESEARCH INTERESTS

- Transportation economics
- Urban & regional development
- Environmental economics



RESEARCH THEMES

1

Infrastructure, Rail networks, regional inequality & post-disaster recovery

- HSR, expressways & post-GEJE revitalization — *Transportation Research A*, 2026
- Sectoral Decarbonization and HSR — *Energy Economics*, 2026 (Top 0.2% Journal)
- Double-edged trains: HSR & regional disparity — *Transport Policy* · 2023

2

ESG, sustainability & financial performance under crises

- Disclosure or Action: ESG behavior & financial performance — *Finance Research Letters* · 2021
- Sustainability performance during financial crises (COVID-19) — *Energy Policy* · 2021

3

Energy economics & environmental policy

- Diesel Policy Abandonment and Consumer Behavior- *Energy Economics*. 2020
- Global mortality benefits of COVID-19 action — *Tech. Forecasting & Social Change* · 2020

4

Micro-mobility, AVs & freight innovation

- Drone vs truck: choice & WTP for last-mile delivery — *Transportation Research E*, 2026
- Ethical Dilemma of Autonomous Vehicle — *Transportation Research Part A*, 2023

교통경제란 무엇인가

- 사람들이 움직이는 것에 대한 경제학
- Core questions: which roads, rails, and airports are worth building; how to price them (congestion charges, fares, fuel taxes); and what they cost society beyond the ticket (congestion, accidents, noise, emissions).
- Its central insight: transport is a DERIVED demand. Nobody flies for the pleasure of sitting in seat 43E; people buy access, and access shapes trade, jobs, and where activity locates.



Transport economics asks what a connection is worth, and who pays its full cost.

And transportation & energy economics?

- The intersection of the two fields: transport runs almost entirely on energy, and mostly on oil.
- It studies how transport choices and infrastructure drive energy use and emissions, and how to decarbonise mobility (fuel taxes, EVs, SAF, modal shift) without losing the access benefits.

 Transport $\approx$ 1/4 of global energy-related CO₂

 Aviation $\approx$ 2–3% of global CO₂, and growing

 And aviation has no near-term electrification path

Today's talk sits exactly at this intersection: a network investment (connectivity) meets an energy outcome (jet fuel burned).

What this paper does

- We compute CO₂ for every scheduled flight in the world, stage by stage, for 6,000+ airports over 1996–2023.
- We then ask a causal question: when a country becomes better connected by air, how much does its aviation CO₂ rise?
- The answer is: much more than proportionally. A 1% gain in connectivity raises national aviation CO₂ by 5.7%.
- We finish with the bill: who owes it, where it is booked, and what (only fuel switching) can shrink it.

A data contribution (the emissions panel) and a causal contribution (the elasticity).

One index, two ledgers: the benefit and the bill

THE BENEFIT (our companion trade paper)

+1.30%

trade openness per 1% rise in hub quality

≈ \$7.8 trillion of 2023 trade, 17% of the world total

THE BILL (this paper)

+5.67%

aviation CO₂ per 1% rise in hub quality

356 Mt of 2023 CO₂, \$18–68 billion per year

Same countries, same connectivity index, same IV logic: the trade paper found the benefit, this paper prices the bill.

Seven parts

01



Motivation

Policy premise meets carbon

02



Data

CO2 for every scheduled flight

03



Building GACI

Five centralities, PCA, one index

04



Identification

A geography instrument

05



Results

The elasticity and its anatomy

06



The carbon bill

Attribution, mismatch, SAF

07



Conclusion

What it implies

01

Motivation

Policy premise meets carbon reality



Everyone wants connectivity; nobody priced the carbon

- Governments subsidise routes, expand airports, and sign air-service agreements on the premise that connectivity raises trade, tourism, and productivity (it does: our companion paper finds +1.3% trade openness per 1% of connectivity).
- But aviation is among the hardest transport sectors to decarbonise: no near-term electrification path, and emissions booked under separate bunker conventions, outside national inventories.
- The direction is obvious (more connections = more flying). The magnitude and structure are not, and no causal estimate exists at the global level.

The carbon consequence of connectivity policy is a first-order question with no causal answer. Until now.

Three empirical questions

01 

Proportionality

Does a 1% gain in connectivity raise emissions by more or less than 1%? Hubs consolidate traffic into fewer, fuller, more efficient movements, so the answer is not obvious.

02 

Incidence

Whose emissions respond? Poor or rich countries? And does the response spill across borders through the network?

03 

Accounting

Are emissions booked where they are generated? International aviation follows bunker conventions, not territory.

Magnitude, geography, and bookkeeping: all three turn out to matter.

02

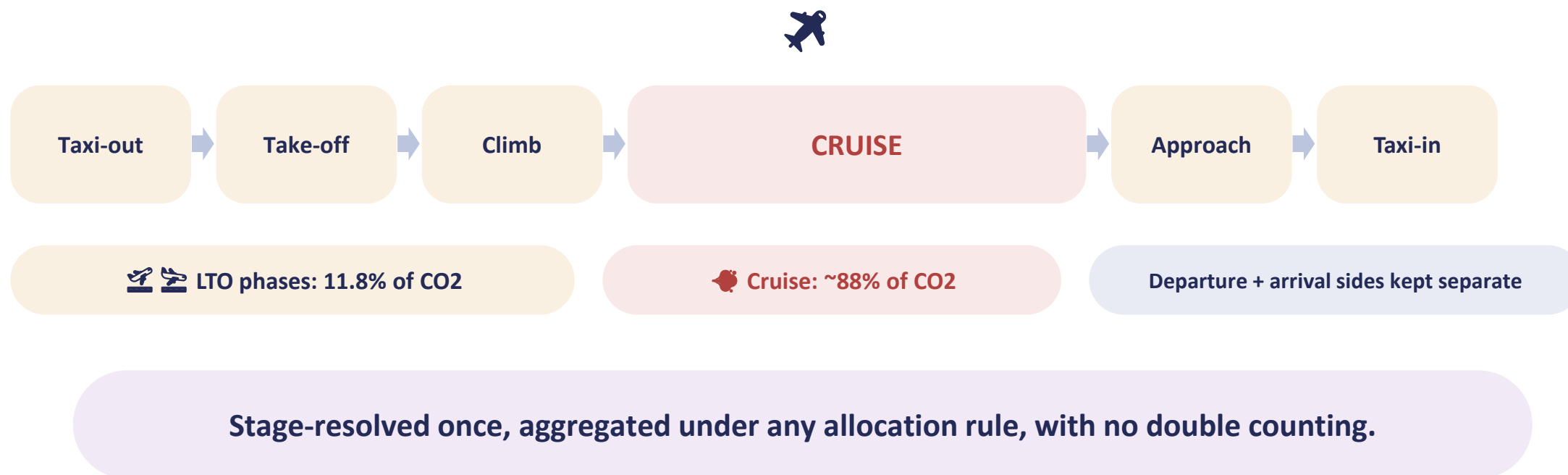
Data

CO2 for every scheduled flight on Earth, 1996–2023



Carbon for every flight, stage by stage

- Source: OAG schedules, every scheduled commercial flight, Jan 1996 to Jun 2024.
- EEA/EMEP Guidebook: fuel burn by engine type × stage length, × 3.15 kg CO₂ per kg of Jet A.



Does it add up? Yes, to the tonne (almost) ☒

0.919 Gt

our 2019 world total vs ICCT's 0.92 Gt

$r = 0.99$

log-log match with OWID/ICCT country series,
every year

11.8%

LTO share of fuel burn, right in the standard
range

6,000+

airports with stage-resolved monthly CO₂

184

countries in the estimation panel

4,634

country-years, 1996–2023

Schedule-based capacity, not load factors: the carbon content of the network itself.

Whose emissions are they? Three answers, one dataset



Territorial (LTO)

Each landing-and-take-off phase booked at its own airport. Physical, local.



Bunker (headline)

★ our headline

Cruise assigned to the departure country: the IEA fuel-uplift convention.

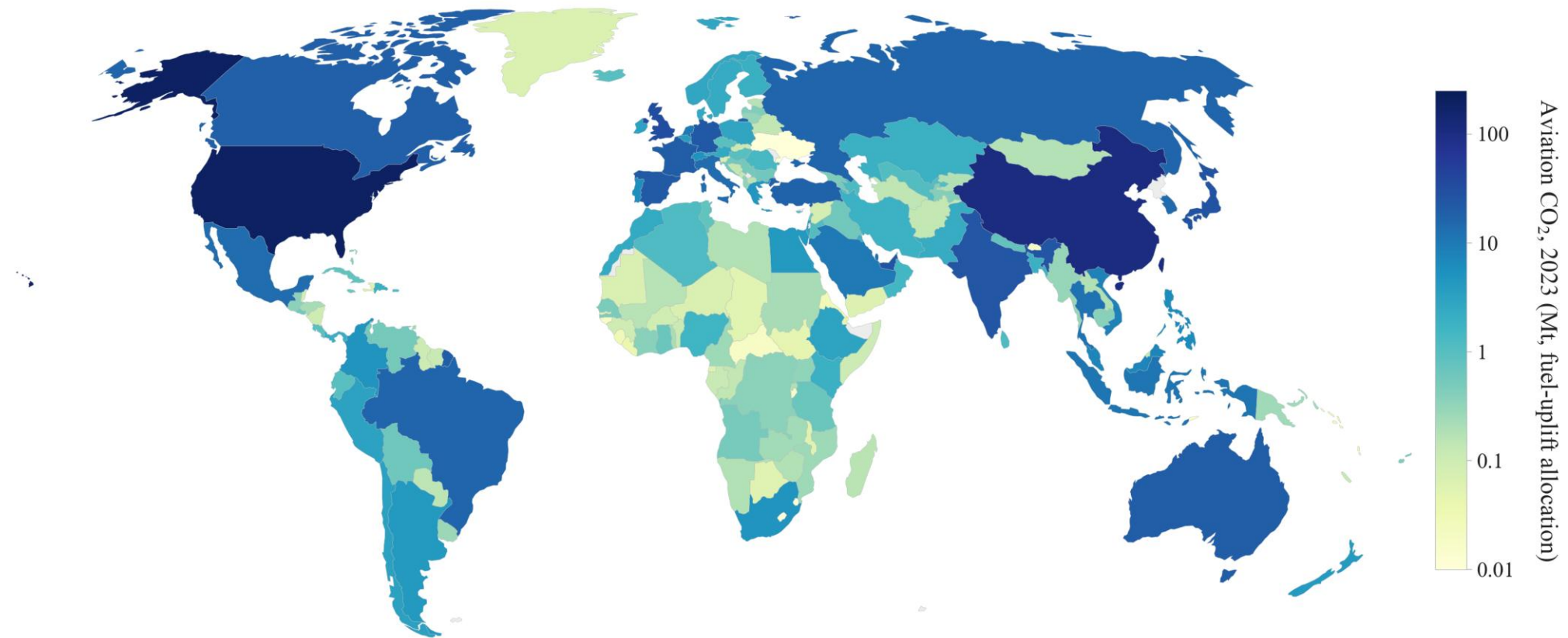


50/50 split

Cruise shared equally between the endpoint countries.

Spoiler: the elasticity barely moves across all three (5.62–5.92). The result is not an accounting artefact.

Where aviation CO₂ lives, 2023



National aviation CO₂ under the bunker convention, 2023. This is the outcome variable.

The emitters: top 15 countries, 2023

#	Country	ISO	CO2 2023 (Mt)	World share	Intl share	LTO (Mt)	Rank 1996	CO2 1996 (Mt)
1	United States	USA	193.4	23.1%	37%	21.4	1	180.1
2	China	CHN	103.9	12.4%	19%	15.7	8	10.3
3	United Kingdom	GBR	32.9	3.9%	96%	2.9	3	18.6
4	Japan	JPN	26.4	3.1%	59%	3.7	2	28.4
5	UAE	ARE	25.7	3.1%	100%	1.7	20	4.7
6	India	IND	24.3	2.9%	48%	3.3	19	5.6
7	Germany	DEU	22.9	2.7%	96%	2.2	4	16.0
8	Spain	ESP	21.9	2.6%	80%	2.8	12	7.1
9	Australia	AUS	20.0	2.4%	60%	2.0	7	11.3
10	France	FRA	19.2	2.3%	91%	2.0	5	12.5
11	Canada	CAN	18.4	2.2%	64%	1.8	6	12.1
12	Turkey	TUR	17.4	2.1%	81%	2.1	32	1.8
13	Brazil	BRA	16.0	1.9%	41%	2.2	10	8.5
14	Russia	RUS	15.8	1.9%	31%	1.7	9	10.1
15	Korea	KOR	14.4	1.7%	90%	1.4	13	7.1

Reading the list

838 Mt of world aviation CO2 in 2023 under the bunker convention.

US + China = 35% of the world total. China: 8th in 1996, 2nd today.

Movers: UAE 20th to 5th, Turkey 32nd to 12th, India 19th to 6th, Qatar into the top 20.

Korea: 15th, 14.4 Mt, 90% international.

Intl share = international share of bunker CO2; LTO = physical landing-and-take-off CO2 at own airports.

Aviation CO2 is concentrated, and its geography has shifted toward the newly connecting world since 1996.

03

Building GACI

What the index is, and exactly how we compute it



What is "air connectivity", really?

- It is not how many airports or routes a country has.
- It is a country's POSITION in the world air network: how easily, through how many and how important partners, it can reach everywhere else.
- Two airports with 'international' in the name are not equal: Incheon plugs into the world's hub network; Yangyang has a handful of point-to-point routes.

Connectivity = network position, not a simple count.

How we measure it: the Global Air Connectivity Index

- GACI summarises each airport's position in the weighted world air network as a single, time-comparable number (Cheung, Wong & Zhang, 2020).
- It blends three ideas:

REACH: how many, and how well-placed, are its connections

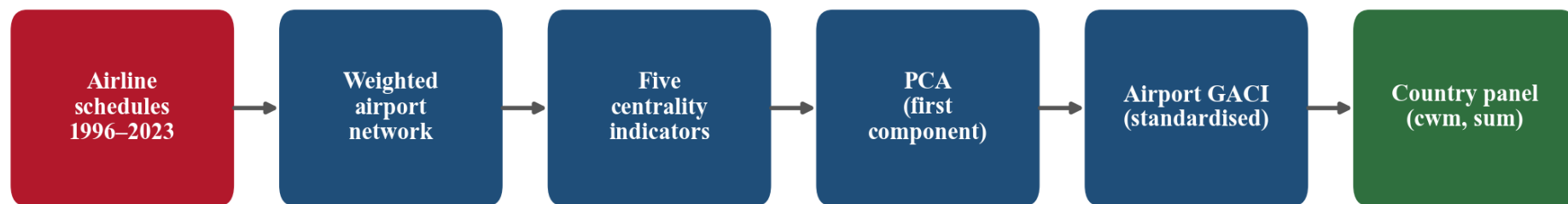
FLOW: how much passenger traffic moves through it

STANDING: how important are the airports it connects to

One number that captures reach, flow, and standing in the global network.

How the index is built: the pipeline

From raw schedules to one number per airport per year:



BUILD THE INDEX (this paper's data contribution)

USE FOR TRADE

today: USE FOR CO2 ✈

Each step is computed for every year, then aggregated to countries.

The raw data: OAG airline schedules

CarrDom1	CarrDom1Name	Origin	Origin.1	DepCity	DepCityName	DeplATACT	DeplATACT	FstSeats	BusSeats	EcoSeats	Frequency	Seats
US	USA	LAX	Los Angeles	LAX	Los Angeles	US	USA	0	0	175	1	175
NO	Norway	TOS	Tromsø	TOS	Tromsø	NO	Norway	0	0	39	1	39
US	USA	SAT	San Antonio	SAT	San Antonio	US	USA	0	12	114	1	126
CN	China	CTU	Chengdu (CTU)	CTU	Chengdu (CTU)	CN	China	0	30	271	1	301
US	USA	BOG	Bogotá	BOG	Bogotá	CO	Colombia	0	0	0	1	0
NZ	New Zealand	PER	Perth	PER	Perth	AU	Australia	0	12	162	1	174
CN	China	SZX	Shenzhen (SZX)	SZX	Shenzhen (SZX)	CN	China	8	0	153	1	161
IN	India	FRA	Frankfurt (FRA)	FRA	Frankfurt (FRA)	DE	Germany	0	24	144	1	168
DE	Germany	LYS	Lyon (LYS)	LYS	Lyon (LYS)	FR	France	0	28	126	1	154
US	USA	SAN	San Diego	SAN	San Diego	US	USA	0	16	144	1	160
US	USA	PHL	Philadelphia	PHL	Philadelphia	US	USA	0	20	170	1	190
MX	Mexico	QRO	Querétaro	QRO	Querétaro	MX	Mexico	0	0	186	1	186
TZ	Tanzania United Republic of	SEU	Seronera	SEU	Seronera	TZ	Tanzania U	0	0	13	1	13
US	USA	DAL	Dallas Love	DFW	Dallas	US	USA	0	0	143	1	143
US	USA	BNE	Brisbane	BNE	Brisbane	AU	Australia	0	8	168	1	176
QA	Qatar	CKY	Conakry	CKY	Conakry	GN	Guinea	0	12	147	1	159
CN	China	PKX	Beijing Dax	BJS	Beijing	CN	China	0	12	147	1	159
CN	China	CAN	Guangzhou	CAN	Guangzhou	CN	China	0	17	162	1	179
CN	China	XIY	Xi'an Xian	SIA	Xi'an	CN	China	0	0	0	1	0
GB	United Kingdom	MIA	Miami Inter	MIA	Miami	US	USA	0	20	214	1	234
KR	Korea Republic of	YHZ	Halifax Star	YHZ	Halifax	CA	Canada	0	12	124	1	136
US	USA	LIR	Guanacaste	LIR	Guanacaste	CR	Costa Rica	0	8	120	1	128
ES	Spain	LPA	Gran Canaria	LPA	Gran Canaria	ES	Spain	0	0	70	1	70
QA	Qatar	RDU	Raleigh/Dur	RDU	Raleigh/Dur	US	USA	0	12	64	1	76
US	USA	HND	Tokyo Intl	TYO	Tokyo	JP	Japan	5	42	205	1	252

Connected dummy
Total seat capacity (the link weight)

Route-segment level, every scheduled commercial flight, 1996–2024H1.

Step 0: building a clean weighted network

- Source: global airline schedule data at the route-segment level (OAG; it is big).
- Keep only scheduled commercial service; exclude military, charter, seaplanes, heliports, and airports that closed in-window.
- Aggregate segments per airport pair per year into one link; the link weight is passenger seat capacity.
- Two matrices per year: adjacency A (who connects) and weight W (how much).

Adjacency A: who connects

	ICN	GMP	CJU	PUS	YNY
ICN	–	0	1	0	0
GMP	0	–	1	1	0
CJU	1	1	–	1	1
PUS	0	1	1	–	0
YNY	0	0	1	0	–

Weight W: seat capacity (millions, illustrative)

	ICN	GMP	CJU	PUS	YNY
ICN	–	0	0.5	0	0
GMP	0	–	17.0	4.0	0
CJU	0.5	17.0	–	2.5	0.1
PUS	0	4.0	2.5	–	0
YNY	0	0	0.1	0	–

The Korean corner of the matrix: Gimpo–Jeju alone is the world's busiest air route.

Step 1a: topological indicators (position)

- Degree: the number of direct connections (raw reach).
- Closeness: inverse of the average number of transfers needed to reach all other airports (accessibility).

Toy domestic network: GMP–CJU, GMP–PUS, GMP–YNY, CJU–PUS

Hops	GMP	CJU	PUS	YNY	Avg
GMP	–	1	1	1	1.00
CJU	1	–	1	2	1.33
PUS	1	1	–	2	1.33
YNY	1	2	2	–	1.67

Airport	Closeness	Reading
GMP	1.00	Hub: everyone in one hop
CJU	0.75	Intermediate
PUS	0.75	Intermediate
YNY	0.60	Spoke: always via Gimpo

+ Eigenvector: how much am I connected to "BIG" airports?

Airport	Degree (# friends)	Eigenvector (rank of friends)
GMP	3	High
CJU	2	High
PUS	2	High
YNY	1	Low

Yangyang's one friend is a big hub, but one friend is still one friend.

Step 1b: volumetric indicators (passenger flow)

- Flow betweenness: the share of passengers transferring through the airport; the hub / transfer role (Gimpo < Incheon: transfers concentrate at ICN).
- Regional importance: average connection intensity within its own region (Gimpo > Incheon: the domestic backbone).
- These two flow measures make GACI flow-aware, not purely topological.

Position says where you sit; flow says how much actually moves through you.

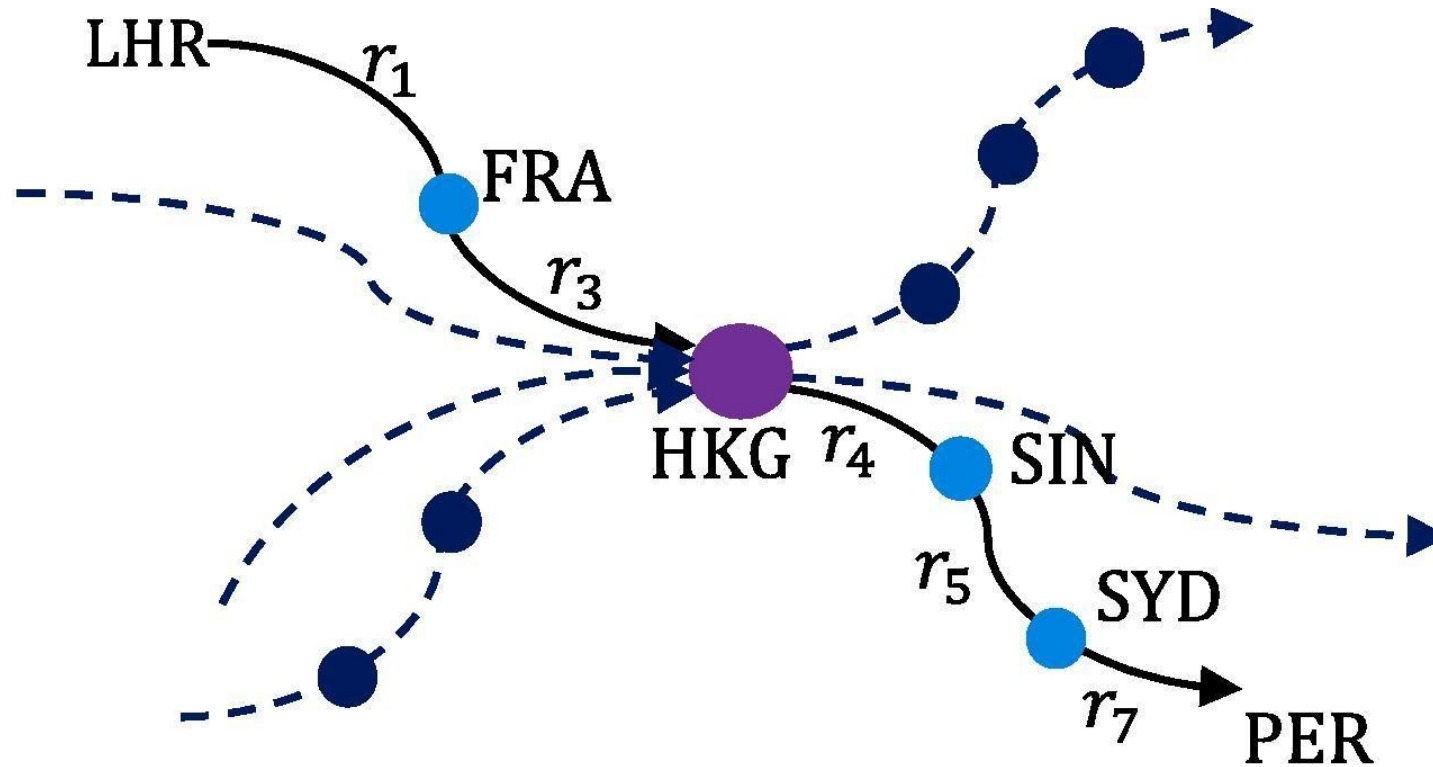
The five indicators, read through Korean airports

Airport	Degree	Closeness	Eigenvector	Flow betw.	Regional imp.
ICN Incheon	High	High	High	High	Mid
GMP Gimpo	Mid	Mid	Mid	Low	High
CJU Jeju	Mid	Mid	Low	Low	High
PUS Gimhae	Mid	Mid	Mid	Low	Mid
YNY Yangyang	Low	Low	Low	Low	Low

ICN: global gateway, strong on every margin. GMP and CJU: enormous traffic (the world's busiest route) but little global position: volume is not connectivity. YNY: connected on paper only.

Five airports, five different network stories: no single indicator captures them all, so we combine all five.

Flow betweenness: passengers as electrical current



Each route is a resistor with resistance inversely proportional to its passenger intensity; betweenness = current through the node. Incheon plays this HKG-style bridge role between North America and Southeast Asia. Source: Cheung et al. (2020).

Step 1: the exact formulas

Step 1: five airport-level centrality indicators

Built on the weighted airport network $G(V, E)$, link weight w_{ij} = passenger seat capacity

Degree

$$d_i = \sum_j a_{ij} \quad (\text{number of direct connections})$$

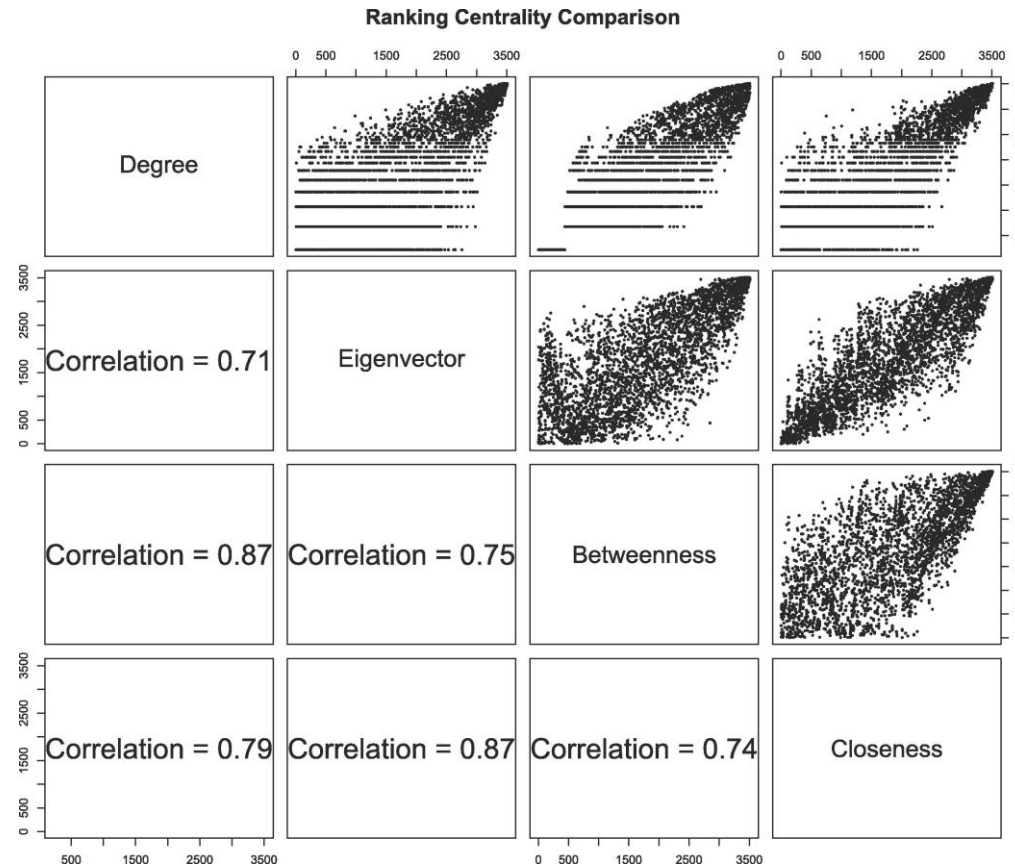
Flow betweenness (hub / transfer role)

$$F_b(i) = \frac{1}{N_B} \sum_{j \neq g} \tau_{jg}(i), \quad \text{resistance} \propto 1/\bar{w}_{ij}, \quad \bar{w}_{ij} = \sqrt{(s_i/d_i)(s_j/d_j)}$$

Closeness, Eigenvector, Regional importance

$$C_c(i) = \frac{N-1}{\sum_j \text{Dist}(i,j)}, \quad C_E(i) = \frac{1}{\lambda} \sum_j W_{ij} C_E(j), \quad C_R(i) = \bar{w}_{ij}^{\text{region}}$$

The five indicators are related but not redundant



Pairwise rank correlations 0.71–0.87: each adds information, so we combine rather than pick one. Source: Cheung et al. (2020).

Step 2: combine the five into one index (PCA)

- Five indicators, different scales, partly correlated.
- GACI = the first principal component: the single weighted combination capturing the most variance across airports.
- On our 1996–2023 panel, PC1 explains 74% of total variance, so one number retains most of the information.

One index, five ingredients, no arbitrary weights.

Step 2: the aggregation formula

Step 2: combine into one index by PCA

Each indicator is standardised (comparable within year and across years)

Composite index

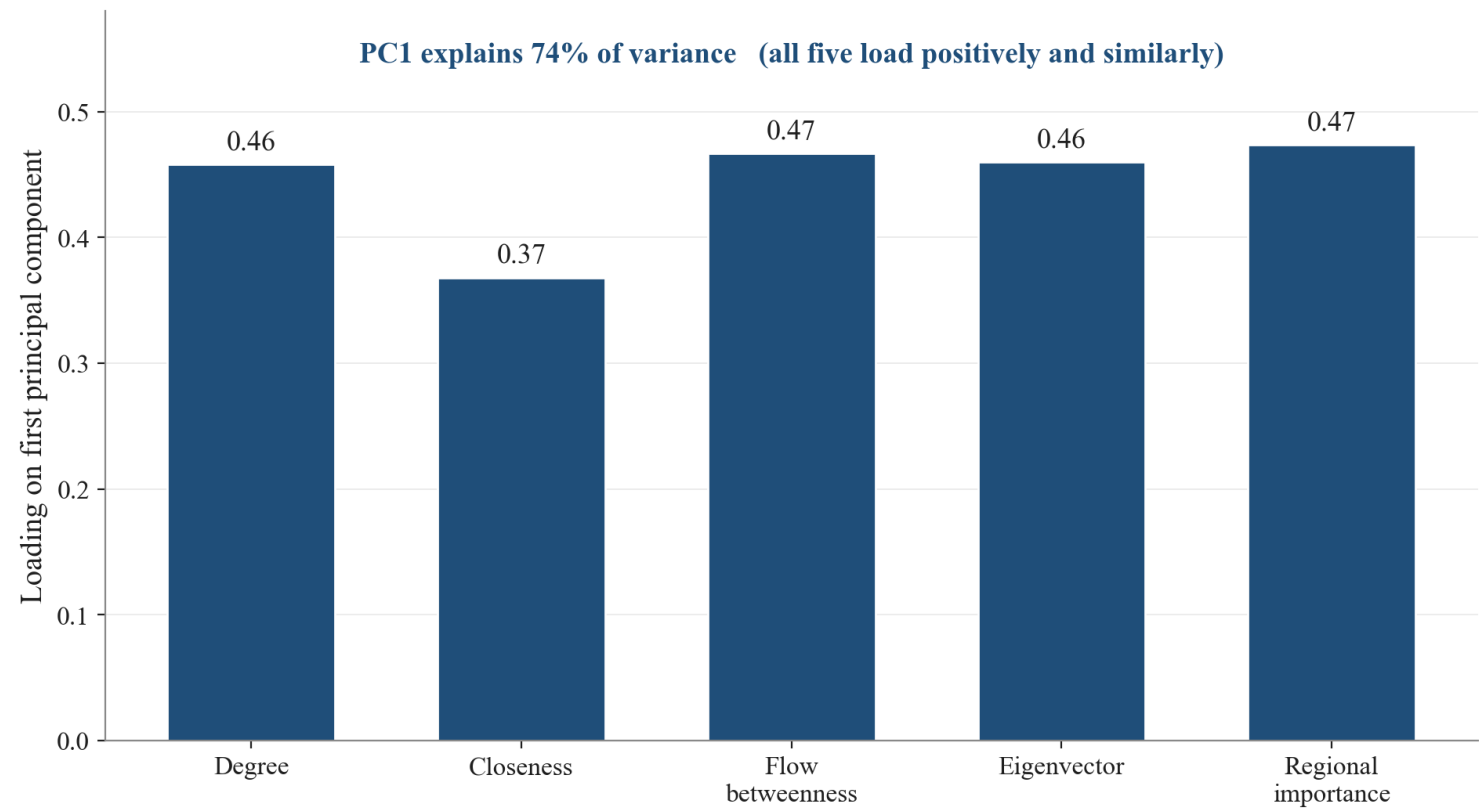
$$\text{GACI} = [D, C_c, F_b, C_E, C_R] \omega$$

ω = first principal component (direction of maximum variance)

The first component explains 65--68% of total variance

Result: an airport-level, time-comparable connectivity score

PCA weights, computed on our panel



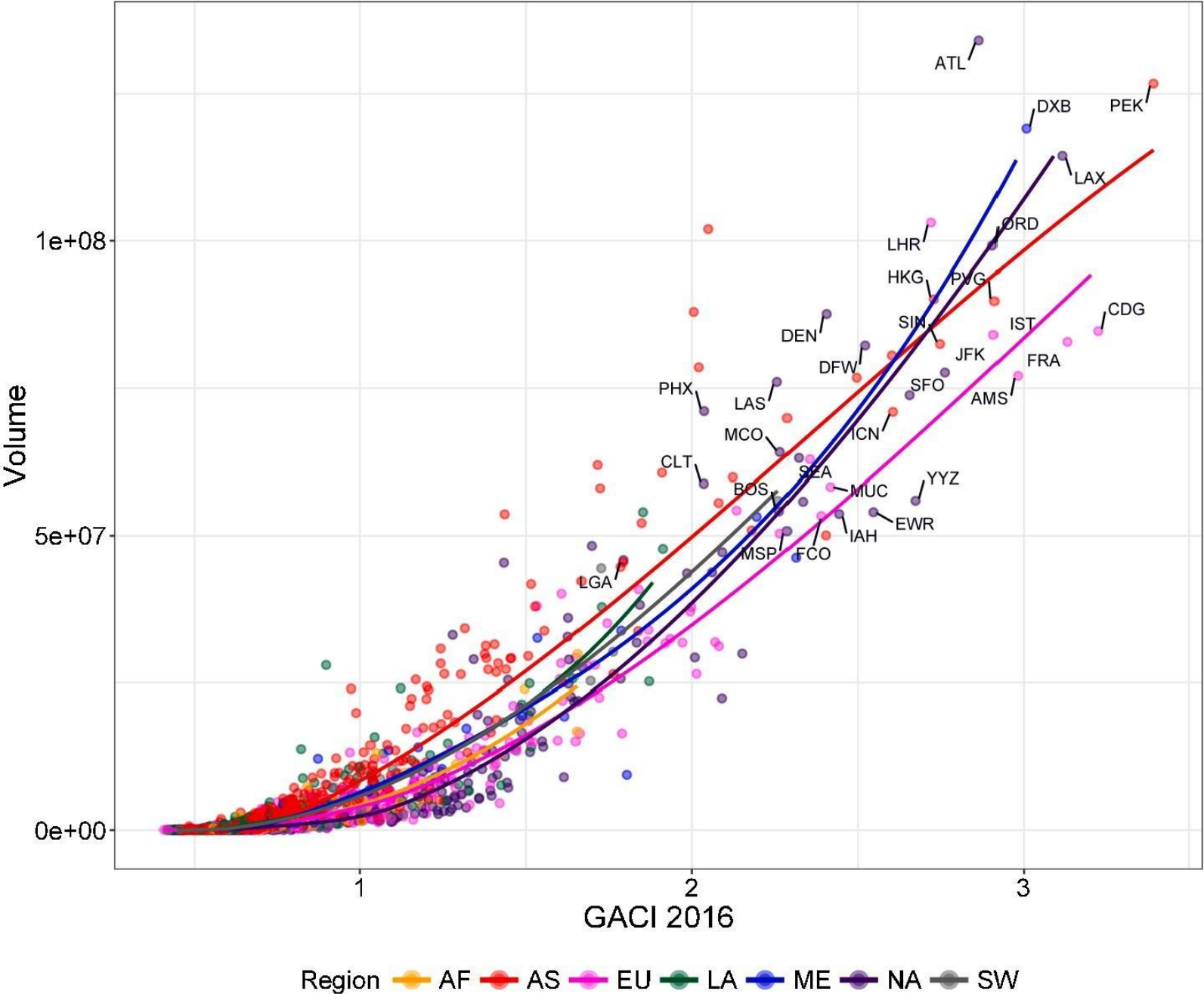
First-component loadings on our 1996–2023 data: all positive and similar; PC1 explains 74% of variance.

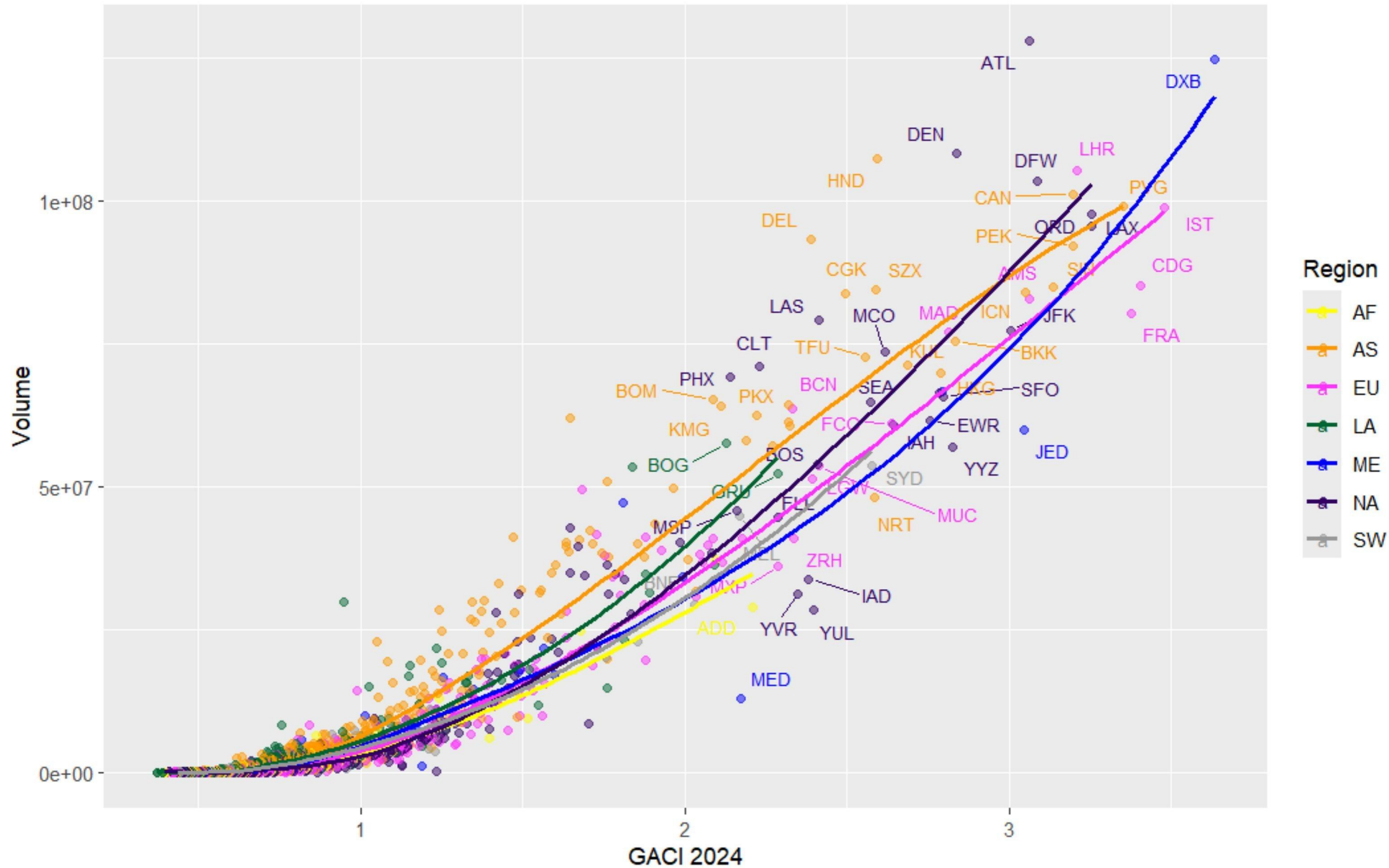
Making GACI comparable over time

- A naive yearly PCA would re-scale every year, so scores could not be compared across years.
- We standardise so that rankings WITHIN a year and relative scores ACROSS years are both preserved.
- The result: one index that tracks an airport or country up and down the global hierarchy from 1996 to 2023.

Essential here: the CO2 question is about changes over 28 years.

Face validity:
high GACI = the world's major hubs





Airport ranking, 2023: the top 15 (of 3,833 scored)

#	Airport	Country	GACI 2023	Seats (M)	Rank 1996
1	FRA	Germany	3.24	76	2
2	IST	Turkey	3.24	95	98
3	DXB	UAE	3.23	117	92
4	CDG	France	3.13	83	4
5	LAX	United States	3.11	96	1
6	ORD	United States	3.06	91	3
7	LHR	United Kingdom	3.04	101	9
8	ATL	United States	3.04	129	5
9	AMS	Netherlands	2.97	78	10
10	DFW	United States	2.94	97	7
11	DEN	United States	2.84	113	16
12	JED	Saudi Arabia	2.82	54	89
13	JFK	United States	2.79	77	6
14	PVG	China	2.77	80	new
15	MIA	United States	2.76	63	8

Korean airports in the same ranking

#	Airport	GACI	Seats (M)	1996
24	ICN	2.60	68.1	new
293	PUS	1.18	15.6	347
320	CJU	1.16	26.1	319
500	GMP	0.96	22.2	new
708	TAE	0.84	2.7	724
877	CJJ	0.79	3.2	new

ICN sits in the world top 25: Korea's one global gateway, opened in 2001 and already above NRT and HND in reach.

GMP and CJU carry the world's busiest route yet rank 300-500: enormous domestic volume, little network reach.

The top of the table is FRA, IST, DXB: Istanbul and Dubai were ranked in the 90s in 1996.

GACI ranks network position: the Gulf and Turkey climbed from the 90s into the top 3 in one generation.

Step 3: from airports to countries

- Hub quality = capacity-weighted MEAN (headline): average gateway quality, not inflated by having many tiny airports.
- Total connectivity = SUM: the overall network footprint, larger for big countries.
- In this paper the headline treatment is hub quality; sum, max, and unweighted mean all reappear as robustness checks (spoiler: they agree).

In GACI (capacity-weighted mean) is the right-hand side of everything that follows.

Step 3: the aggregation formulas

From airports to countries

Two national summaries of the airport-level GACI

Hub quality (capacity-weighted mean) -- headline

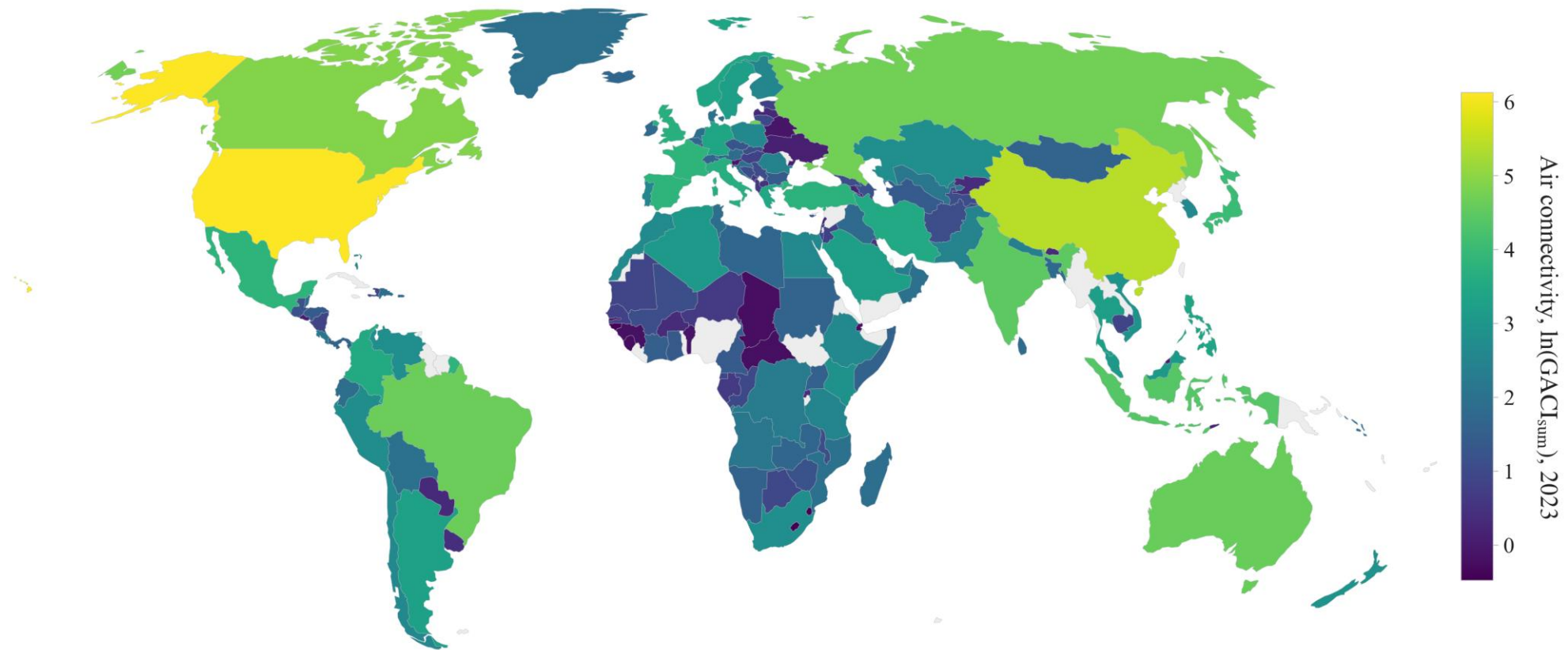
$$\ln \text{GACI}_c^{\text{cwm}} = \ln \left(\frac{\sum_{a \in c} \text{cap}_a \text{GACI}_a}{\sum_{a \in c} \text{cap}_a} \right)$$

Total connectivity (country sum)

$$\ln \text{GACI}_c^{\text{sum}} = \ln \left(\sum_{a \in c} \text{GACI}_a \right)$$

Panel: 184 countries x 1996--2023 = 4,643 country-years

The built index, mapped: connectivity by country, 2023



Total connectivity (country sum), 2023. Yellow = highly connected.

Country ranking, 2023: hub quality vs network size

Hub quality: capacity-weighted mean GACI (the regressor), 166 countries

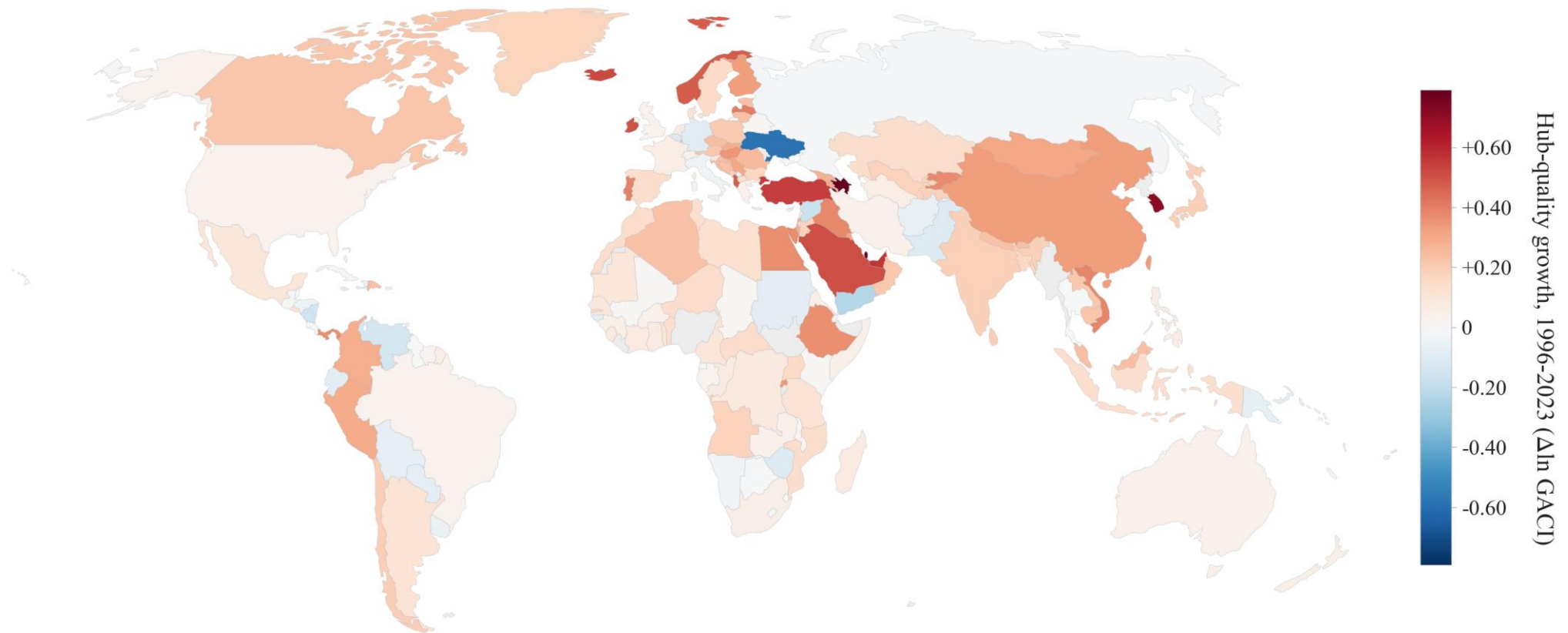
#	Country	Hub quality	Best airport	Airports
1	UAE	2.79	3.23	7
2	Netherlands	2.74	2.97	5
3	Singapore	2.67	2.68	2
4	Hong Kong	2.15	2.15	1
5	Germany	2.15	3.24	29
6	United States	2.09	3.11	569
7	Turkey	2.08	3.24	50
8	France	2.06	3.13	53
9	United Kingdom	2.06	3.04	42
10	Switzerland	1.99	2.21	4
11	Saudi Arabia	1.97	2.82	28
12	Israel	1.90	1.93	2
13	Ethiopia	1.87	2.08	19
14	Austria	1.87	1.99	6
15	Ireland	1.84	2.02	6
19	Korea	1.79	2.60	16

Network size: sum of airport GACI (an alternative aggregation)

#	Country	GACI sum	Airports
1	United States	459	569
2	China	226	254
3	Canada	135	211
4	Russia	113	174
5	Brazil	103	166
6	Australia	101	160
7	India	89	117
8	Indonesia	80	123
9	Japan	56	70
10	France	47	54
11	Spain	46	48
12	Mexico	45	57
13	Turkey	44	50
14	United Kingdom	41	42
15	Italy	39	39
44	Korea	14	16

The regression uses hub quality: small hub states (UAE, Netherlands, Singapore) lead; the size list is a different world. Korea: 19th vs 44th.

The treatment: who got connected, 1996–2023



Change in log connectivity, 1996–2023. The Gulf, Turkey, China, and Korea connect; parts of Europe quietly disconnect.

04

Identification

Breaking reverse causality with geography



The problem: airlines follow the economy

- Airlines add capacity where income, trade, and travel demand are already growing: reverse causality, upward and downward biases at once.
- OLS would also be attenuated by measurement error in the index (it is: OLS = 3.59 vs 2SLS = 5.67).
- We need variation in connectivity that has nothing to do with a country's economic trajectory.

Our answer: geography that was fixed before the boom, interacted with a global technology cycle.

The model: a log-log panel with two-way fixed effects

$$\ln \text{CO}_{2,ct} = \beta \ln \text{GACI}_{ct} + \gamma \ln \text{pop}_{ct} + \delta \ln \text{seaMA}_{ct} + \alpha_c + \tau_t + \varepsilon_{ct} \quad (1)$$

β = elasticity of national aviation CO2 with respect to hub quality: the % change in CO2 per 1% change in GACI. Benchmark: $\beta = 1$ (emissions grow in proportion to connectivity).

Variables

$\text{CO}_{2,ct}$ national aviation CO2 in year t; headline = bunker rule. Also LTO-only, 50/50 split, international only, seat-km, CO2 per seat-km.

GACI_{ct} hub quality: capacity-weighted mean of airport GACI scores (alternatives: max, mean, sum).

pop_{ct} population (log).

seaMA_{ct} sea-distance market access (log): the surface-shipping alternative, held fixed.

α_c country fixed effects: geography, size, institutions.

τ_t year fixed effects: fuel prices, world traffic cycle, technology.

ε_{ct} error; robust standard errors.

Why OLS is not enough

Reverse causality: airlines add capacity where demand is already growing.

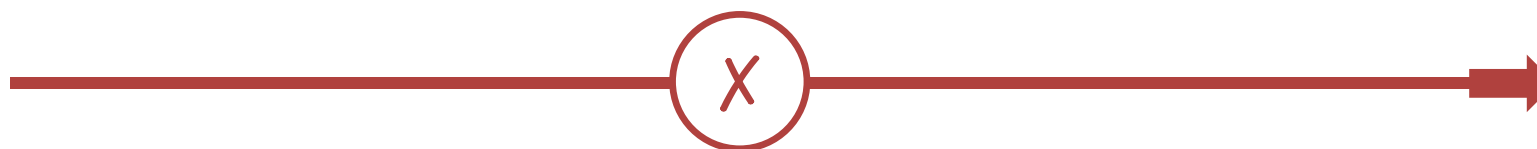
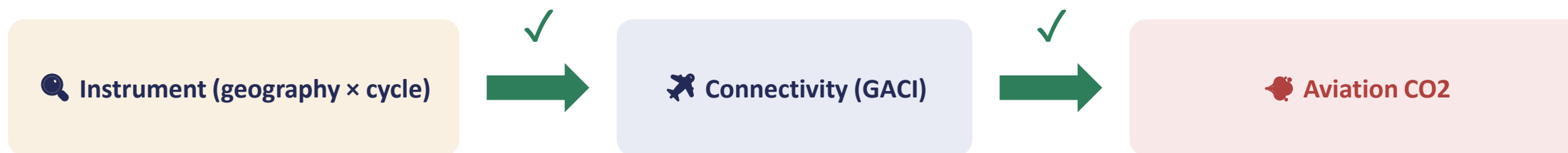
Measurement: the index is an estimate; noise attenuates OLS.

Sample: 184 countries × 1996–2023, N = 4,634 country-years.

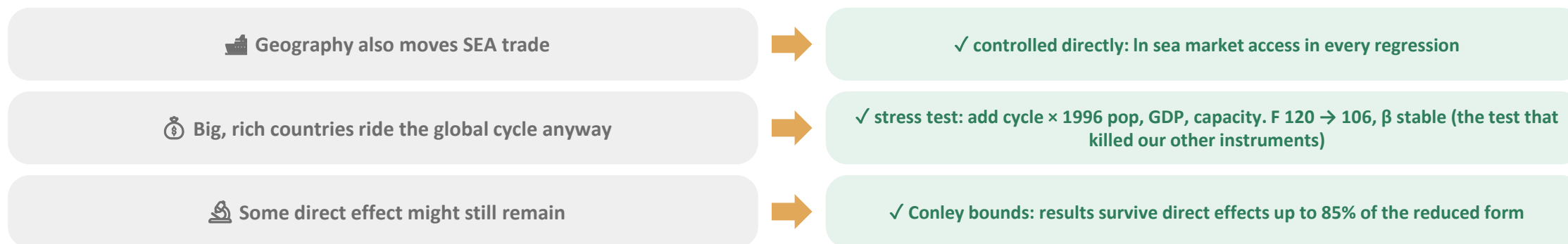
Answer: instrument $\ln \text{GACI}$ with fixed 1996 geography × the world aviation cycle (next slides). OLS gives 3.59; 2SLS gives 5.67.

The fine print: the exclusion restriction

The instrument may affect CO2 through ONE door only: connectivity.



no direct path to CO2 allowed (untestable by assumption, so we attack it three ways)



A Feyrer-type air-vs-sea geography instrument



Controls: log population, log sea market access, country and year fixed effects.

Kleibergen–Paap F = 153.6

Stress test (cycle × baseline size): F 120 → 106. Tourism IV: 14 → 0, dies.

Conley bounds: tolerates direct effects up to 85% of the reduced form

The only instrument we tried that survives its own execution squad.

Two-stage least squares, step by step

1

First stage: predict connectivity from geography

$$\ln \text{GACI}_{ct} = \pi Z_{ct} + \gamma_1 \ln \text{pop}_{ct} + \delta_1 \ln \text{seaMA}_{ct} + \alpha_c + \tau_t + u_{ct}$$

$$Z_{ct} = a_t \times \ln \text{airMA}_{c,1996} \quad (\text{Feyrer-type shifter})$$

a_t : world aviation capacity index, min-max scaled 0 (1996) to 1: the global cycle nobody steers.

$\text{airMA}_{c,1996}$: air market access on fixed 1996 geography, the differential exposure to that cycle.

$\hat{\pi} = 0.128$ (s.e. 0.010), $t = 12.4$, Kleibergen–Paap $F = 153.6$
Reduced form: $\ln \text{CO}_2$ on Z directly = 0.726 (s.e. 0.057).

2

Second stage: plug the prediction into (1)

$$\ln \text{CO}_{2,ct} = \beta \ln \hat{\text{GACI}}_{ct} + \gamma \ln \text{pop}_{ct} + \delta \ln \text{seaMA}_{ct} + \alpha_c + \tau_t + \varepsilon_{ct}$$

Exactly identified: one instrument, one endogenous regressor, so β is a ratio of two reduced forms: $0.726 / 0.128 = 5.67$.

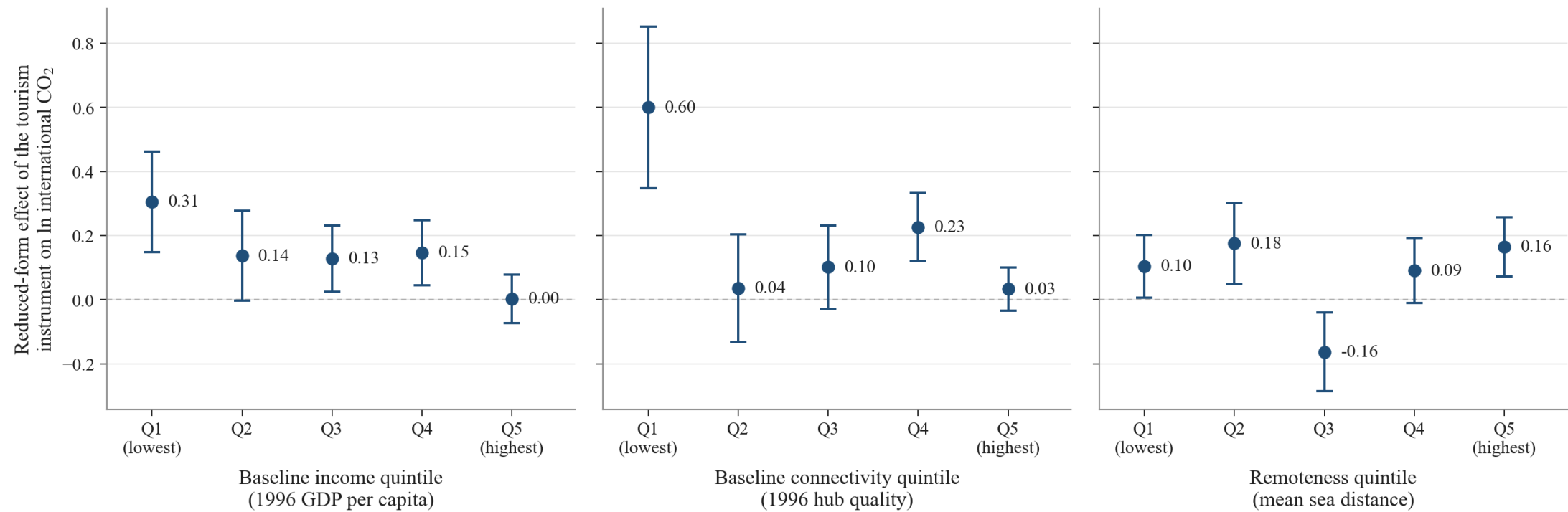
Only the geography-driven part of connectivity growth is used; demand-driven capacity is purged.

Inference: robust SEs; Conley (spatial) SEs and plausibly-exogenous bounds as checks.

$\beta = 5.669$ (s.e. 0.412), $p < 0.001$
OLS on the same sample: 3.585 (0.175). $N = 4,634$.

A 1% rise in hub quality raises national aviation CO2 by 5.7%: far from the proportional benchmark of one.

The raw first look: shifter up, emissions up



Reduced form by instrument quintile: countries with stronger geography-driven connectivity growth emit more.

05

Results

The elasticity, and everything inside it



+1% → +5.7%

connectivity up one percent, national aviation CO2 up 5.67 percent (s.e. 0.41)

The benchmark is an elasticity of one: hubs could in principle absorb growth efficiently.

An elasticity of one is rejected at any conventional level.

Main results: OLS versus 2SLS across six outcomes

Dependent variable (log)	Bunker CO2	LTO CO2	50/50 CO2	Intl. CO2	Seat-km	CO2 / seat-km
Panel A. OLS, country and year FE						
In GACI (cwm)	3.585*** (0.175)	3.043*** (0.177)	3.566*** (0.175)	3.874*** (0.198)	3.941*** (0.183)	−0.355*** (0.043)
Panel B. 2SLS, Feyrer instrument, capacity-weighted mean (headline)						
In GACI (cwm)	5.669*** (0.412)	5.921*** (0.448)	5.620*** (0.409)	5.690*** (0.415)	6.068*** (0.443)	−0.399*** (0.118)
Kleibergen–Paap F	153.6	153.6	153.6	152.7	153.6	153.6
Panel C. 2SLS, alternative aggregations of airport GACI to the country						
In GACI (sum)	2.283*** (0.224)	2.384*** (0.219)	2.263*** (0.222)	2.291*** (0.237)	2.443*** (0.250)	−0.161*** (0.052)
KP F = 90.3						
In GACI (max)	4.822*** (0.329)	5.036*** (0.348)	4.780*** (0.327)	4.836*** (0.343)	5.161*** (0.361)	−0.340*** (0.101)
KP F = 172.0						
In GACI (mean)	5.897*** (0.591)	6.159*** (0.623)	5.846*** (0.587)	5.902*** (0.593)	6.313*** (0.624)	−0.415*** (0.122)
KP F = 136.2						
N (country-years)	4,634	4,634	4,634	4,633	4,634	4,634

Headline

$\beta = 5.67$ (0.41)
+1% hub quality = +5.7% national aviation CO2.
Benchmark of one rejected by a wide margin.

OLS understates
3.59 vs 5.67: the 2SLS estimate exceeds OLS, so the net OLS bias is downward.

Allocation rule irrelevant
5.62 to 5.92 across territorial, bunker and split conventions.

Scale vs technique
Seat-km +6.07 and intensity −0.40: volume, not efficiency.

Notes: country-year panel, 184 countries, 1996–2023. All columns include country and year fixed effects and control for log population and log sea market access. Instrument: world aviation cycle × 1996 air market access. Robust SE in parentheses. ***, **, *: 1, 5, 10% significance.

However you count it: far above one

+5.67***

Bunker CO2 (headline)

+5.92***

LTO-only (territorial)

+5.62***

50/50 cruise split

+5.69***

International CO2

+6.07***

Seat-kilometres

−0.40***

CO2 per seat-km

Aggregation checks: GACI sum +2.28***, max +4.82***, mean +5.90*** (KP F 90–172). 2SLS, Feyrer IV, N = 4,634.

Allocation rule, aggregation, instrument: nothing pushes the elasticity anywhere near one.

Efficiency is real. It is also hopelessly outgunned.

Scale (seat-km)

+6.07%

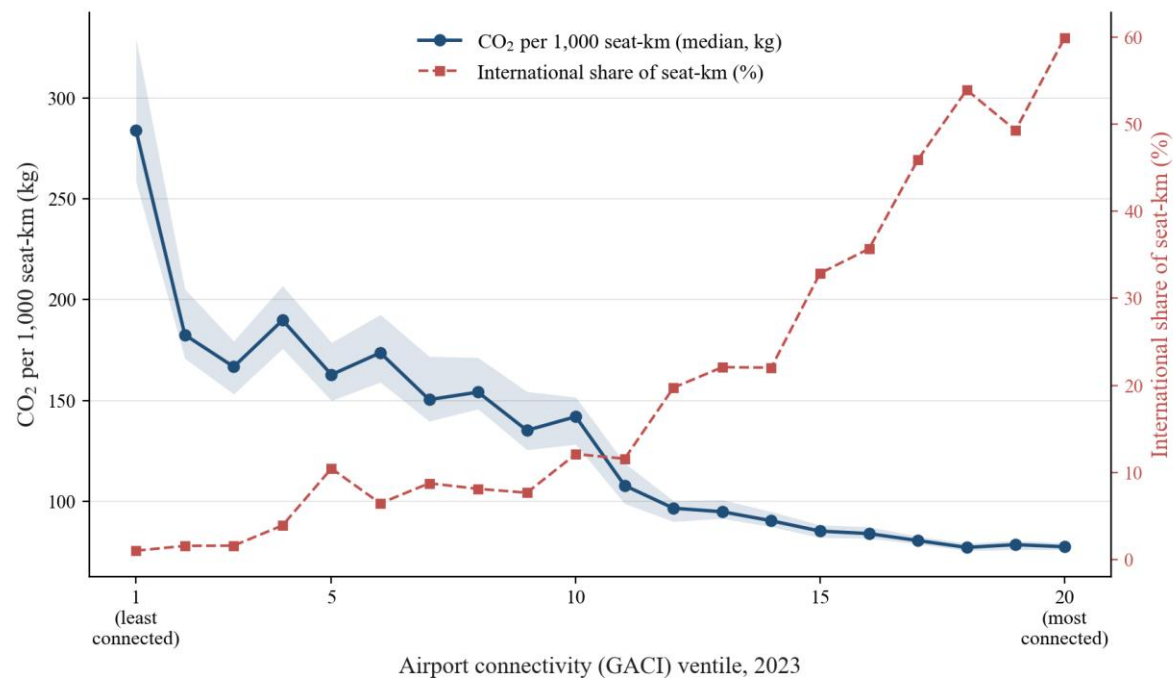
Technique (CO2 per seat-km)

−0.40%

- The same 1% of connectivity raises seat-kilometres by 6.07%, faster than emissions rise.
- So emissions per seat-km fall by 0.40%: hubbing does deliver larger, fuller, longer-range operations.
- But the efficiency margin offsets **less than one tenth of the scale response**.

The emissions consequence of connectivity is a volume story, not an efficiency story.

Zoom to the airport: the efficiency curve

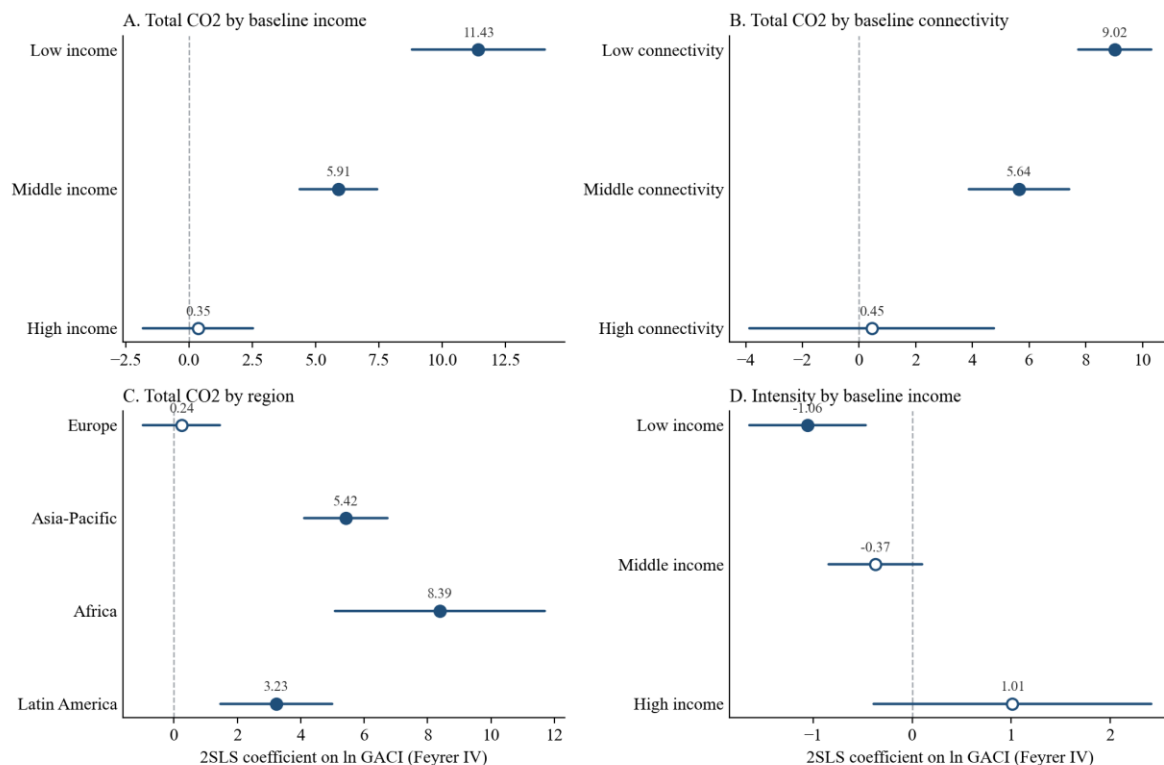


2023 cross-section by connectivity ventile: median intensity falls 284 → 77 kg per 1,000 seat-km while the international share climbs from 1% to 60%.

Efficiency is real at the node, and dominated by scale at the aggregate.

Who drives it: the newly connecting world

Heterogeneity of the connectivity elasticity (filled = $p < 0.05$, bars = 95% CI)



By 1996 income: 11.4 (bottom) · 5.9 (middle) · 0.35 ns (top)

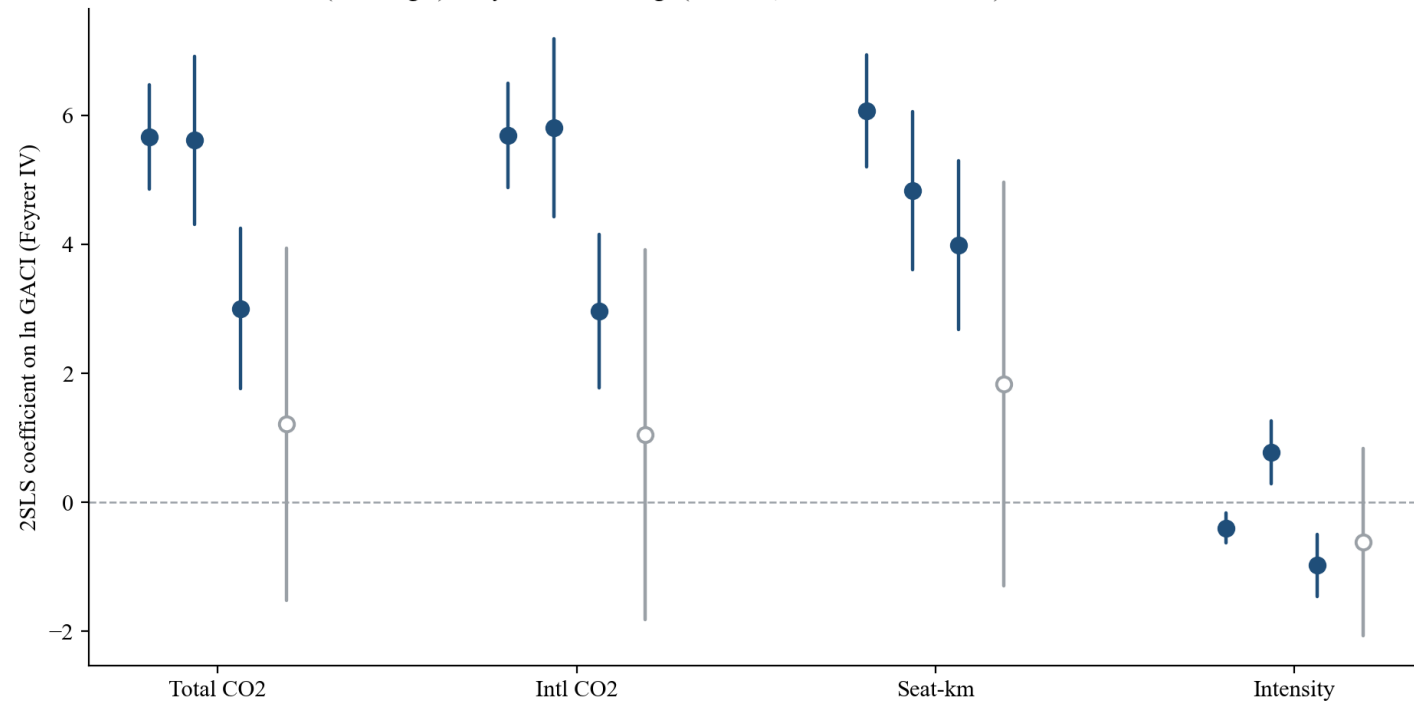
By region: Africa 8.4 · Asia-Pacific 5.4 · Latin America 3.2 · Europe 0.24 ns

Efficiency gains also appear only in the bottom income tercile.

For mature networks, connectivity growth is roughly carbon-neutral at the margin. The elasticity is the sound of countries entering the network.

And it is fading as the network matures

Temporal split: within each outcome, dots = full 1996–2023 / 1996–2007 / 2010–2023 excl. 2020–21 / 2010–2023 unrestricted (left to right). Gray = weak first stage (KP F = 7, COVID-contaminated).



5.61 → 3.01 excluding crisis years: the elasticity halves from 1996–2007 to 2010–2023 ($p = 0.005$).

+0.78 → -0.98 the efficiency margin only appears in the mature era.

COVID alone breaks the late-period first stage (F 36.7 → 7.0 with 2020–21 included).

Still far above one in both eras: maturity slows the meter, it does not stop it.

Who and when: split-sample 2SLS

A. Heterogeneity (total bunker CO2; terciles on 1996 values)

Subsample	β	s.e.	KP F	N
By baseline income				
Low income	11.431***	(1.338)	35.2	1,595
Middle income	5.906***	(0.782)	48.2	1,547
High income	0.350	(1.106)	10.1	1,491
By baseline connectivity				
Low connectivity	9.018***	(0.656)	117.4	1,452
Middle connectivity	5.642***	(0.908)	29.9	1,526
High connectivity	0.452	(2.201)	1.8	1,655
By region				
Europe	0.236	(0.616)	55.5	1,189
Asia-Pacific	5.424***	(0.667)	64.4	1,110
Africa	8.385***	(1.688)	20.2	1,230
Latin America	3.229***	(0.900)	14.1	683

Middle East and North America: not identified (KP F < 2). Robust SE in parentheses; ***, **, *: 1, 5, 10%.

B. Temporal split (2SLS, Feyrer instrument)

Sample	Bunker CO2	Intl. CO2	Seat-km	Intensity	KP F
Full, 1996–2023	5.669***	5.690***	6.068***	−0.399***	153.6
N = 4,634	(0.412)	(0.415)	(0.443)	(0.118)	
Expansion era, 1996–2007	5.613***	5.809***	4.837***	+0.776***	75.0
N = 1,883	(0.665)	(0.704)	(0.626)	(0.249)	
Mature era, 2010–23 ex-COVID	3.005***	2.965***	3.984***	−0.979***	36.7
N = 2,065	(0.635)	(0.610)	(0.668)	(0.248)	
Unrestricted 1996–2009	6.109***	6.262***	5.537***	+0.571***	127.7
N = 2,218	(0.565)	(0.590)	(0.534)	(0.192)	
Unrestricted 2010–2023	1.213	1.051	1.829	−0.617	7.0
N = 2,415 (weak IV)	(1.394)	(1.463)	(1.598)	(0.739)	

Pre vs post difference: 2.61 (p = 0.005). The elasticity halves as the network matures but stays far above one. The weak unrestricted post-2010 first stage is a COVID artefact: dropping 2020–21 restores F to 36.7.

The elasticity is the sound of a network being built: newly connecting, low-income countries carry it; mature networks are near carbon-neutral at the margin.

Anatomy: volume carries it, hub quality recomposes it

What mediates the effect?

86.5%

of the effect runs through seat-kilometres

83.9%

through flight frequency

~0%

through the international share

A different lever, hub quality (secondary tourism IV), moves the mix instead:

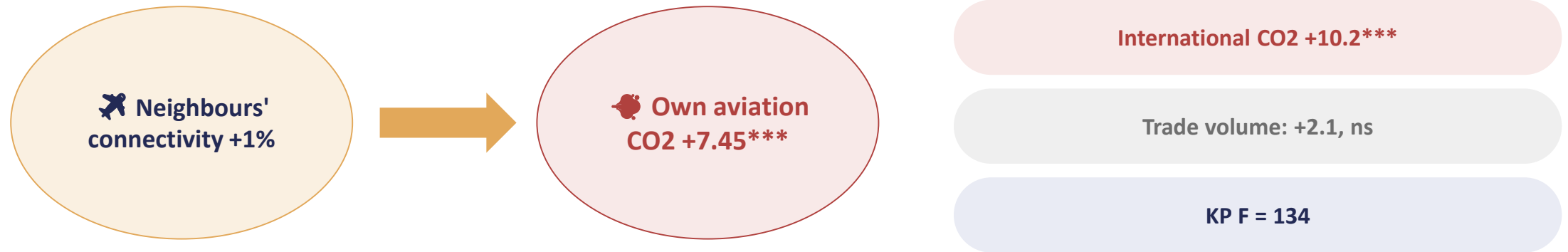
Total CO2: no response

International CO2: +3.16***

CO2 per seat-km: +1.23***

Market access scales flying; hub quality tilts it toward international long-haul.

Your neighbour's hub, your emissions



The margins are the converse of the own-network response:

Flights: ns

Aircraft size +5.4***

Intl share +1.0***

Stage length -2.0**

Feeding foreign hubs: the CO2 lands at home, the trade gains do not. National accounting misses a regional externality.

Spillovers: neighbours' connectivity, own outcomes

Panel A. 2SLS: neighbour connectivity instrumented by neighbours' Feyrer shifters (inverse-distance weighted)

Dependent variable (log)	Bunker CO2	Intl. CO2	CO2 / seat-km	Trade volume
In neighbour GACI	7.454*** (1.994)	10.237*** (2.092)	1.312** (0.534)	2.072 (ns) (2.011)
Own Feyrer shifter (reduced form)	0.254** (0.122)	0.076 (0.128)	-0.134*** (0.037)	0.157 (0.124)
Kleibergen–Paap F	134.1	134.0	134.1	134.1
N	4,623	4,622	4,623	4,623

Panel B. Margins of the own response to neighbour connectivity (same specification)

Dependent variable (log)	Flights	Seat-km	Aircraft size	Stage length	Intl. share
In neighbour GACI	2.730 (ns) (2.216)	6.141*** (2.083)	5.437*** (1.170)	-2.026** (1.008)	1.046*** (0.281)

Notes: country and year FE; controls log population and log sea market access; robust SE in parentheses. ***, **, *: 1, 5, 10% significance. ns = not significant.

How to read it

Neighbour GACI: distance-weighted average of all other countries' log hub quality.

Why reduced form for own: own and neighbour shifters are collinear within country, so only one can be instrumented.

Result: +1% in neighbours' hubs raises own CO2 by 7.5% and international CO2 by 10%, with no trade gain.

Margins: not more flights, but bigger aircraft, shorter stages, more international share: feeding the foreign hub.

Feeding foreign hubs: the CO2 lands at home, the trade gain does not.

06

The carbon bill

Attribution, who pays, and the only real lever



How much did GACI rise, and how much CO2 is that?

1 Connectivity actually rose

Median country: **+8%** GACI 1996–2023; emissions-weighted **+21%**. 43 of 184 countries fell. Korea **+104%** (0.88 → 1.79), UAE +76%, Turkey +73%, China +39%, US +4%.

2 × the elasticity

$\beta = 5.67$: **each +1% GACI = +5.67% CO2**.
Counterfactual with 1996 connectivity: $E_{2023} \times \exp(-\beta \times \Delta \ln \text{GACI})$. Attributed = $E_{2023} \times [1 - \exp(-\beta \Delta \ln \text{GACI})]$.

3 = the carbon bill

World aviation CO2 **457 → 838 Mt** (+83%) over 1996–2023. Attributed to connectivity growth: 356 Mt = 42.5% of 2023 emissions, about 94% of the net increase.

Country	GACI 1996 → 2023	Δ GACI	CO2 1996 → 2023 (Mt)	Δ CO2	Attributed (Mt)	Share of 2023
Korea	0.88 → 1.79	+104%	7.1 → 14.4	+103%	+14.1	98%
China	1.16 → 1.60	+39%	10.3 → 103.9	+911%	+87.7	84%
UAE	1.58 → 2.79	+76%	4.7 → 25.7	+442%	+24.7	96%
Turkey	1.21 → 2.08	+73%	1.8 → 17.4	+841%	+16.6	95%
India	1.18 → 1.42	+21%	5.6 → 24.3	+332%	+15.9	65%
Japan	1.32 → 1.60	+21%	28.4 → 26.4	-7%	+17.4	66%
United States	2.01 → 2.09	+4%	180.1 → 193.4	+7%	+36.4	19%
United Kingdom	1.99 → 2.06	+3%	18.6 → 32.9	+77%	+5.5	17%
Germany	2.36 → 2.15	-9%	16.0 → 22.9	+43%	-15.8	-69%
World (184 countries)	median +8%	wtd +21%	457 → 838	+83%	+356	43%

Read with care

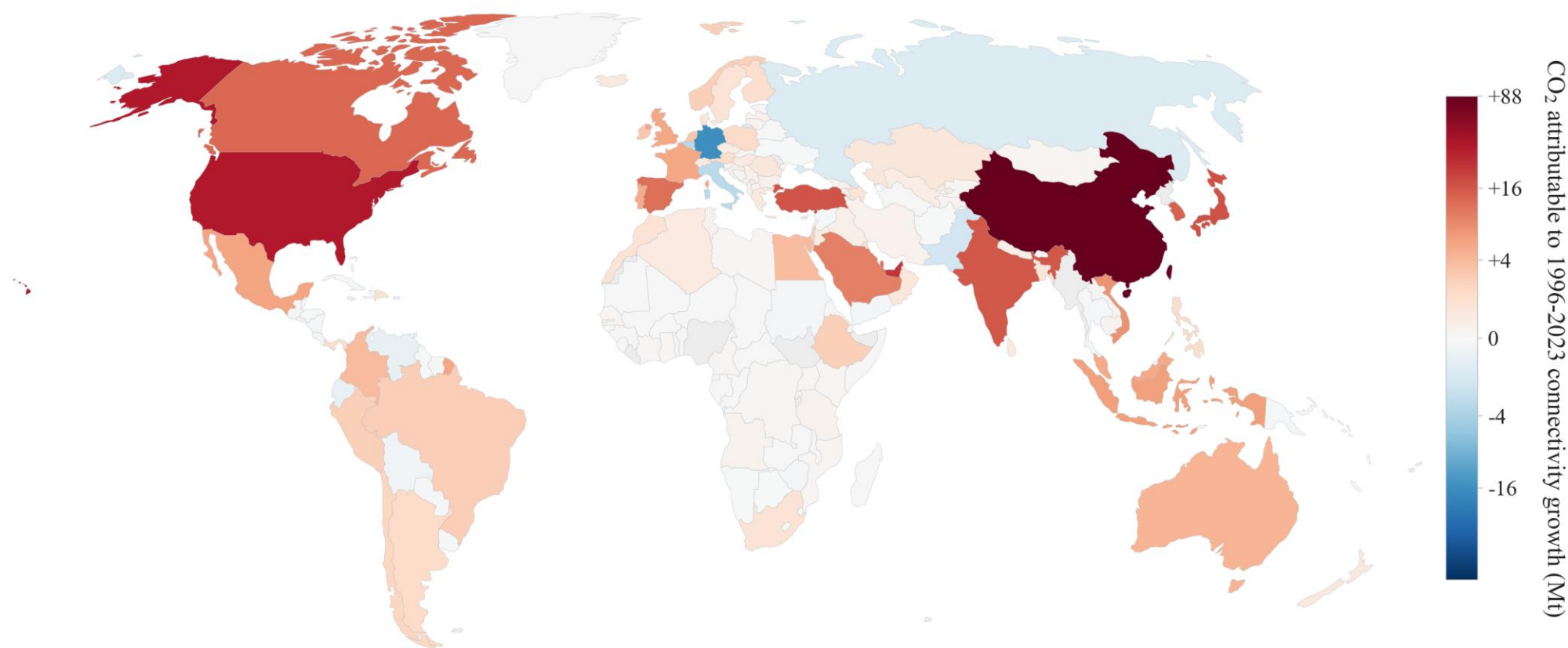
Marginal elasticity: estimated from year-to-year variation, then applied to 28-year changes.

Saturation: the attributed share tends to 100% for big movers (Korea 98%), so country rows are upper bounds.

Negative rows: Germany's GACI fell 9%, so the formula credits it with avoided CO2.

Most countries moved little; the bill is a story of a few dozen fast connectors. Korea doubled its hub quality and doubled its aviation CO2.

356 Mt: the CO₂ the connectivity boom built



2023 aviation CO₂ attributed to post-1996 connectivity growth: 356 Mt, 42.5% of the world total. White = zero.

The price tag: the social cost of carbon

SCC = the present value of all future damage caused by one extra tonne of CO₂.



Three standard price tags (2020 USD per tonne):

\$51 per tonne
US Interagency Working Group (2021) · 3% discount rate

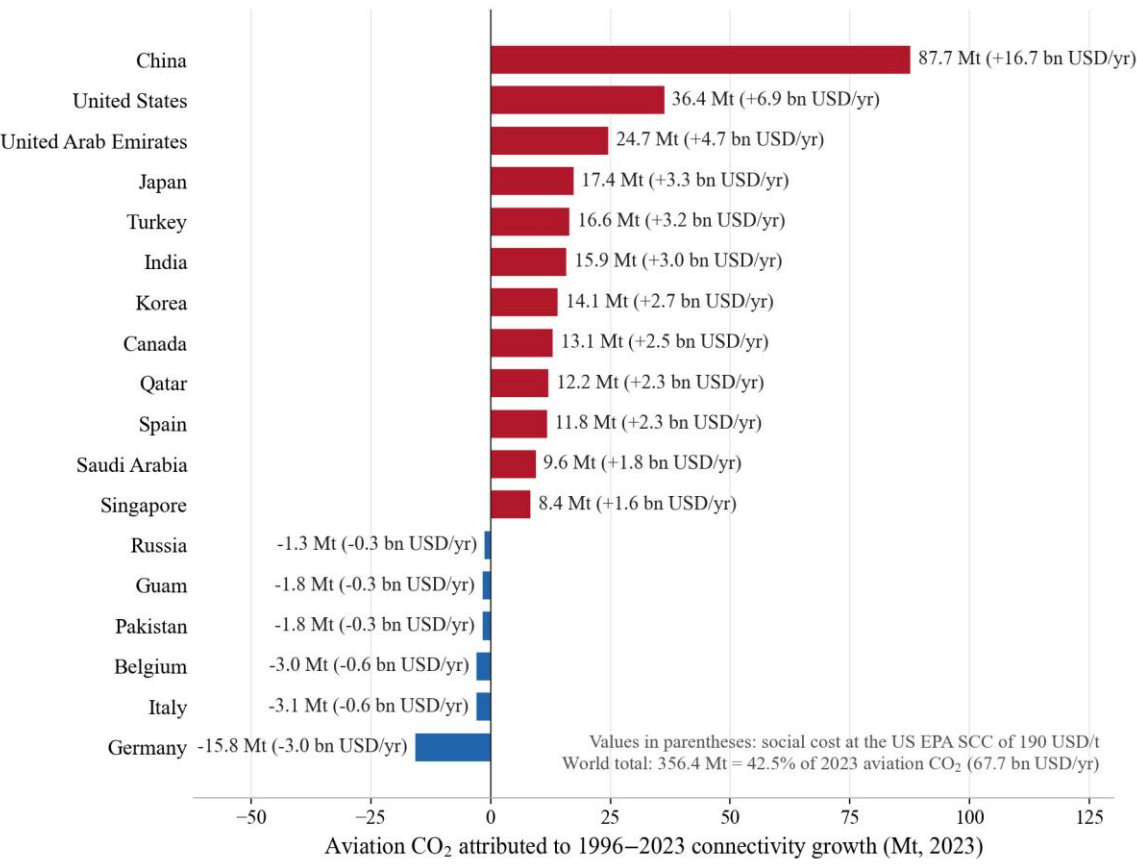
\$185 per tonne
Rennert et al. (2022, Nature) · 2% near-term discount rate

\$190 per tonne
US EPA (2023) central value · 2% Ramsey discounting

The gap is mostly the discount rate, not the climate science: how much do we care about damages 50 years out?

Attach this price tag to our 356 Mt: an \$18–68 billion bill, recurring every year the traffic persists.

Who owes what: the country bill, priced

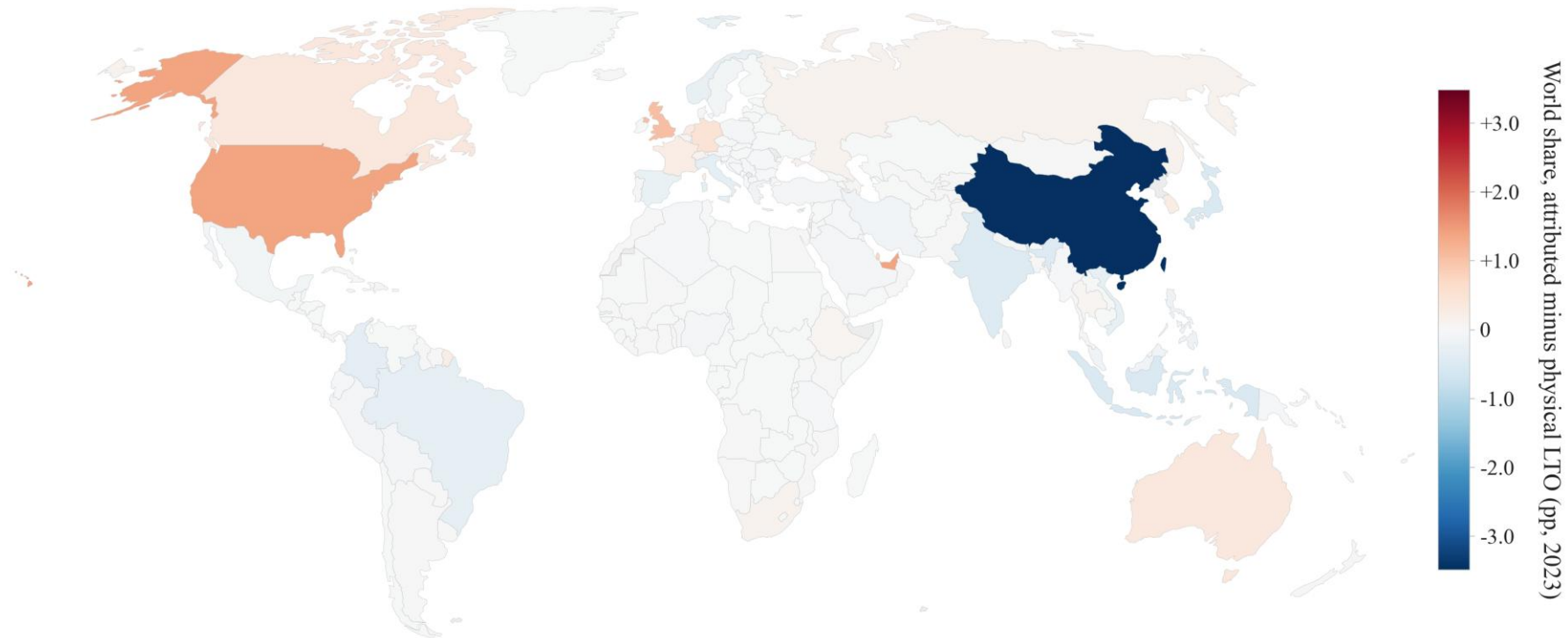


China +88 Mt · \$16.7 bn/yr
United States +36 Mt · \$6.9 bn/yr
UAE +25 Mt · \$4.7 bn/yr
Japan / Turkey +17 Mt each
Korea +14 Mt
Germany -16 Mt · -\$3.0 bn/yr

World: \$18.2 / 65.9 / 67.7 bn per year at SCC \$51 / \$185 / \$190.

A climate bill in the tens of billions of dollars, recurring annually while the traffic persists.

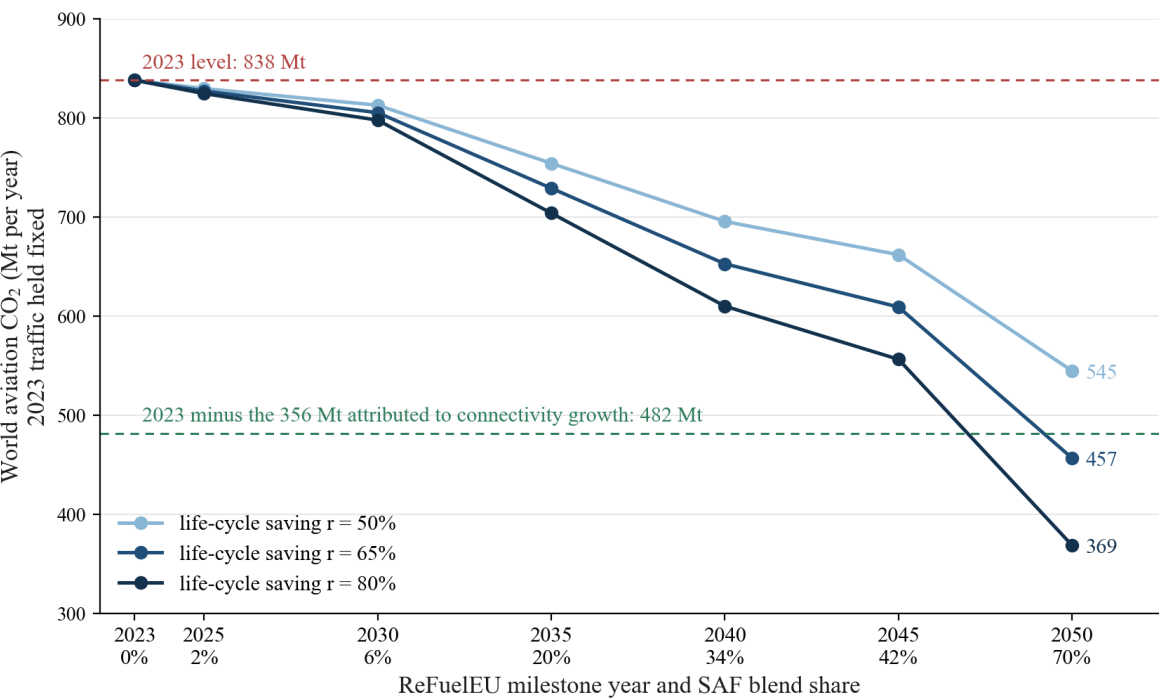
Generated here, booked there



Bunker-attributed share minus physical LTO share, 2023: +1.4 pp for the US and the UAE, -3.5 pp for China. Positive at every international mega-hub.

Any scheme that allocates responsibility by booked emissions, CORSIA included, embeds a transfer across these lines.

The only lever that moves the level: change the fuel



HOW THE SCENARIO IS COMPUTED

$E(s, r) = 838 \text{ Mt} \times [1 - s \times r]$

838 Mt = world aviation CO2 in 2023 (bunker convention); 2023 scheduled traffic held fixed, fuel backed out at 3.15 kg CO2 per kg Jet A.
s = SAF blend share on the ReFuelEU path: 2% (2025), 6% (2030), 20% (2035), 34% (2040), 42% (2045), 70% (2050).
r = life-cycle CO2 saving of SAF vs fossil Jet A: 50 / 65 / 80% (literature range).
g per \$ = $E(s, r) \div$ attributed trade (\$7.8 tn): 21 g/\$ in 2023, 9–14 g/\$ on the 2050 path.

Mt CO2 per year	SAF blend	r = 50%	r = 65%	r = 80%
2023 actual	0%	838	838	838
2035 mandate	20%	754	729	704
2050 mandate	70%	545	457	369
CO2 saved in 2050		293 Mt	381 Mt	469 Mt
... as % of the 356 Mt		82%	107%	132%

World aviation CO2 along the ReFuelEU blending path, 2023 traffic held fixed: 838 Mt → 704–754 Mt at the 2035 mandate → 369–545 Mt at the 2050 mandate. Green line: the 2023 level minus the 356 Mt attributed to post-1996 connectivity growth (elasticity 5.67).

The full 2050 path (≥65% life-cycle saving) offsets about as much CO2 as the past generation of connectivity growth created.

07

Conclusion

Three myths, three implications, one bill



Three comfortable myths, audited

MYTH

WHAT THE DATA SAY

"Hub efficiency will absorb the traffic growth."



It offsets less than one tenth of the scale response.

"Emissions are booked where they are generated."



Hub accounting shifts up to 3.5 pp of world emissions across borders.

"Your emissions depend on your own network."



+1% in neighbours' connectivity raises your CO2 by 7.5%.

Connectivity policy needs a carbon line in its cost-benefit table. Here is the number to put there.

Summary and policy

- New data: flight-stage CO₂ for 6,000+ airports, 1996–2023, valid under any allocation rule.
- New estimate: the connectivity-emissions elasticity is 5.67, far above one, concentrated in newly connecting countries, spilling across borders.
- The bill: 42.5% of 2023 aviation CO₂ (356 Mt, \$18–68 bn per year) traces to the post-1996 connectivity boom.
- Policy: evaluate connectivity investments jointly with fuel and demand-side instruments; efficiency gains are not a decarbonisation strategy, and booked-emissions schemes embed cross-border transfers.

Connect if it pays, but put the carbon in the ledger, and bring the fuel.

Points to discuss (a.k.a. our limitations)

- Schedule-based emissions: carbon content of scheduled capacity, not of realised load factors.
- The instrument is strongest in the expansion era; the late period is identified only excluding COVID years.
- Airport-level results are descriptive: no valid airport-level instrument survived our checks.
- Attribution and SAF scenarios are partial-equilibrium accounting; non-CO2 climate effects (contrails, NOx) are out of scope.
- Where next: load factors? Airline-level behaviour? Non-CO2 forcing? Your ideas welcome.

Thank you

How Much Does Air Connectivity Increase Aviation CO2?

+1% connectivity → +5.7% CO2 · 356 Mt since 1996 · only fuel switching bends the level